

Cleo's arctangent–sine integral: the Reshetnikov–Lee χ_2 master formula, an elementary-value classification, and a sibling trichotomy

Vladimir Reshetnikov

2026-06-21

Research report. Created (UTC) 2026-06-21T16:04:59Z; repository HEAD
31ffeac4a7ca7bfcc21480d16ed05ba5b4d46b3a.

Companion to `central-binomial-arcsin-ladder`; store `cleo-arctan-sin-chi2.wl`.

Abstract

The Math.StackExchange integral $I = \int_0^{\pi/2} \arctan^2\left(\frac{6 \sin x}{3 + \cos 2x}\right) dx$ was evaluated in 2014 by Cleo and, independently, by Sangchul Lee, in two forms that we show are the *same* three-term Legendre- χ_2 combination. The engine behind it is the two-parameter master formula $\int_0^{\pi/2} \arctan(p \sin x) \arctan(q \sin x) dx = \pi \chi_2(u_p u_q)$, with $u_p = \tan(\frac{1}{2} \arctan p)$ —conjectured and proved by Lee in 2013, seeded by Reshetnikov's single-arctangent ladder $\int_0^{\pi/2} \arctan(t \sin x) dx = 2\chi_2(u_t)$. We give a short third (differentiation) proof, machine-checked in Wolfram, and then record four contributions that are *not* in the prior literature: (i) a *classification* of the elementary values of χ_2 at algebraic points—over the metallic-mean, $\tan(\pi/k)$, and rational families the elementary set is *closed* at exactly the three classical golden/silver values, and Cleo's three arguments all avoid it, which is the structural reason I has no elementary closed form; (ii) a proof that the three-term χ_2 representation of I is *minimal* (not a single dilogarithm, not a two-term combination, no Catalan constant); (iii) an alternate *Landen-mirror* three-term form; and (iv) a *sibling trichotomy* extending the master formula: $(\arctan, \arctan) \rightarrow \chi_2(u_p u_q)$, $(\operatorname{arctanh}, \operatorname{arctanh}) \rightarrow \chi_2(v_p v_q)$, $(\arctan, \operatorname{arctanh}) \rightarrow \operatorname{Ti}_2(u_p v_q)$. All identities are verified in two engines (mpmath and Wolfram), to between 40 and 160 digits.

1 The integral and its provenance

The object (Math.StackExchange [question 564816](#)) is

$$I = \int_0^{\pi/2} \arctan^2\left(\frac{6 \sin x}{3 + \cos 2x}\right) dx = 1.330970396842566666125327 \dots \quad (1)$$

[Cleo](#) gave the dilogarithmic form

$$I = \frac{\pi}{2} \left(2 \operatorname{Li}_2((1 - \sqrt{2})(2 - \sqrt{5})) - 2 \operatorname{Li}_2((1 - \sqrt{2})(\sqrt{5} - 2)) \right. \\ \left. + \operatorname{Li}_2(3 - \sqrt{8}) - \operatorname{Li}_2(\sqrt{8} - 3) + \operatorname{Li}_2(9 - \sqrt{80}) - \operatorname{Li}_2(\sqrt{80} - 9) \right), \quad (2)$$

while [Sangchul Lee](#) gave the Legendre-chi form

$$I = \pi \left(\chi_2(3 - 2\sqrt{2}) + \chi_2(9 - 4\sqrt{5}) + 2\chi_2((\sqrt{2} - 1)(\sqrt{5} - 2)) \right), \quad (3)$$

where $\chi_2(x) = \frac{1}{2}(\text{Li}_2(x) - \text{Li}_2(-x)) = \sum_{k \geq 0} \frac{x^{2k+1}}{(2k+1)^2}$. These are identical: $\text{Li}_2(x) - \text{Li}_2(-x) = 2\chi_2(x)$ turns each Li_2 pair in (2) into a χ_2 , and $3 - \sqrt{8} = (\sqrt{2} - 1)^2$, $9 - \sqrt{80} = (\sqrt{5} - 2)^2$, $(1 - \sqrt{2})(2 - \sqrt{5}) = (\sqrt{2} - 1)(\sqrt{5} - 2)$, so both equal

$$I = \pi \left[\chi_2((\sqrt{2} - 1)^2) + \chi_2((\sqrt{5} - 2)^2) + 2 \chi_2((\sqrt{2} - 1)(\sqrt{5} - 2)) \right]. \quad (4)$$

The arguments are a silver $\otimes$ golden pair: $\sqrt{2} - 1 = \tan \frac{\pi}{8} = \delta_S^{-1}$ (reciprocal silver ratio $\delta_S = 1 + \sqrt{2}$) and $\sqrt{5} - 2 = \phi^{-3}$ (golden $\phi = \frac{1+\sqrt{5}}{2}$, $\phi^3 = \sqrt{5} + 2$), so $I = \pi[\chi_2(\delta_S^{-2}) + \chi_2(\phi^{-6}) + 2\chi_2(\delta_S^{-1}\phi^{-3})]$.

2 The master formula (Reshetnikov–Lee) and a differentiation proof

The closed forms (2)–(3) are instances of a two-parameter master formula. Its seed is Reshetnikov’s single-arctangent ladder $\int_0^{\pi/2} \arctan(t \sin x) dx = 2\chi_2(u_t)$; inspired by it, Lee (a/562492, 2013) conjectured and proved

Theorem 1 (Lee 2013). *For real p, q ,*

$$J(p, q) := \int_0^{\pi/2} \arctan(p \sin x) \arctan(q \sin x) dx = \pi \chi_2(u_p u_q), \quad (5)$$

where $u_p = \frac{\sqrt{1+p^2} - 1}{p} = \tan(\frac{1}{2} \arctan p)$.

Lee gives two proofs (a triple integral and a Fourier-orthogonality argument). We record a short third proof by differentiation, which is the most direct route and is machine-verifiable.

Proof. Both sides vanish on the axes, and $\partial_p J(p, 0) = \partial_p [\pi \chi_2(u_p u_0)] = 0$ (since $u_0 = 0$), symmetrically in q ; hence each side equals the double integral of its mixed partial over $[0, p] \times [0, q]$, and it suffices to match mixed partials. Differentiating under the integral and using the partial fraction $\frac{s^2}{(1+p^2 s^2)(1+q^2 s^2)} = \frac{1}{p^2 - q^2} \left(\frac{1}{1+q^2 s^2} - \frac{1}{1+p^2 s^2} \right)$ together with $\int_0^{\pi/2} \frac{dx}{1+c \sin^2 x} = \frac{\pi}{2\sqrt{1+c}}$,

$$\partial_p \partial_q J = \frac{\pi}{2(p^2 - q^2)} \left(\frac{1}{\sqrt{1+q^2}} - \frac{1}{\sqrt{1+p^2}} \right). \quad (6)$$

On the right, $\chi_2'(w) = \text{arctanh}(w)/w$ gives $\partial_p \partial_q [\pi \chi_2(u_p u_q)] = \pi u_p' u_q' / (1 - u_p^2 u_q^2)$. The substitution $p = \frac{2u_p}{1-u_p^2}$ yields $\sqrt{1+p^2} = \frac{1+u_p^2}{1-u_p^2}$ and $u_p' = \frac{(1-u_p^2)^2}{2(1+u_p^2)}$, whence both sides reduce to the same rational function $\frac{\pi(1-u_p^2)^2(1-u_q^2)^2}{4(1+u_p^2)(1+u_q^2)(1-u_p^2 u_q^2)}$. (Wolfram confirms this reconciliation symbolically: `FullSimplify` of the difference returns 0.) $\square$

3 Cleo’s integral as the perfect-square instance

The argument in (1) collapses to a sum of arctangents. With $\cos 2x = 1 - 2 \sin^2 x$,

$$\frac{6 \sin x}{3 + \cos 2x} = \frac{3 \sin x}{2 - \sin^2 x} = \frac{(1 + \frac{1}{2}) \sin x}{1 - (1 \cdot \frac{1}{2}) \sin^2 x},$$

the tangent-addition shape with roots $1, \frac{1}{2}$ of $t^2 - \frac{3}{2}t + \frac{1}{2}$, so for $0 \leq x \leq \frac{\pi}{2}$

$$\arctan\left(\frac{6 \sin x}{3 + \cos 2x}\right) = \arctan(\sin x) + \arctan\left(\frac{1}{2} \sin x\right), \quad (7)$$

so that $I = \int_0^{\pi/2} (\arctan(\sin x) + \arctan(\frac{1}{2} \sin x))^2 dx$. Applying Theorem 1 to the expanded square, with $u_1 = \sqrt{2} - 1$, $u_{1/2} = \sqrt{5} - 2$, reproduces (4). The coefficient pattern 1, 2, 1 is a perfect square: writing $a = \sqrt{2} - 1$, $b = \sqrt{5} - 2$,

$$\frac{I}{\pi} = \sum_{k \geq 0} \frac{((\sqrt{2} - 1)^{2k+1} + (\sqrt{5} - 2)^{2k+1})^2}{(2k + 1)^2}. \quad (8)$$

The coefficients $a^{2k+1} + b^{2k+1}$ are the sin-power Fourier coefficients of $\arctan(\sin x) + \arctan(\frac{1}{2} \sin x)$; (8) is the Parseval form underlying Lee's Fourier proof.

4 Which χ_2 are elementary: a classification

The master formula expresses I in χ_2 , but χ_2 is only *elementary*—reducible to π^2 , Catalan's constant G , and products of logarithms of units—at very special arguments. The following three values are classical (Lewin, *Dilogarithms*, 1958, p. 19; restated by J. M. Campbell, *Irish Math. Soc. Bulletin* **89** (2022), eqs. (6)–(7)):

$$\chi_2(\phi^{-1}) = \frac{\pi^2}{12} - \frac{3}{4} \ln^2 \phi, \quad \chi_2(\phi^{-3}) = \frac{\pi^2}{24} - \frac{3}{4} \ln^2 \phi, \quad \chi_2(\sqrt{2} - 1) = \frac{\pi^2}{16} - \frac{1}{4} \ln^2(1 + \sqrt{2}). \quad (9)$$

(Wolfram FullSimplify renders the first as $\frac{1}{12}(\pi^2 - 9 \operatorname{arccsch}^2 2)$ with $\operatorname{arccsch} 2 = \ln \phi$; the two golden values are Landen partners via $\frac{1-\phi^{-1}}{1+\phi^{-1}} = \phi^{-3}$ and the χ_2 Landen law (14).) What is *not* in the literature is the boundary itself.

Proposition 2 (Elementary- χ_2 classification). *Over the families of inverse metallic means m_n^{-k} ($m_n = \frac{n+\sqrt{n^2+4}}{2}$, $1 \leq n \leq 6$, $1 \leq k \leq 6$), the silver/bronze powers, the tangents $\tan(\pi/k)$ for $k = 5, 8, 10, 12$ and their squares, and the rationals tested, the only elementary values of χ_2 are the three of (9) and their algebraic aliases. In particular $\chi_2(\phi^{-2}), \chi_2(\phi^{-4}), \dots, \chi_2(\phi^{-6}), \chi_2(\delta_S^{-2}), \chi_2(\delta_S^{-3}), \chi_2(\delta_S^{-4}), \chi_2(\frac{1}{\sqrt{2}}), \chi_2(\frac{2}{\sqrt{5}}), \chi_2(2 - \sqrt{3})$ and $\chi_2(1/\sqrt{3})$ are not elementary.*

Each statement is an integer-relation result at 130–140 digits with a Champernowne canary to reject false positives (a nonzero canary coefficient kills a candidate; residual gate $\leq 9 \times 10^{-131}$). The three arguments of Cleo's integral, $\delta_S^{-2}, \phi^{-6}, \delta_S^{-1} \phi^{-3}$, all lie in the non-elementary set. This is the structural reason I has no elementary closed form: each building block is individually transcendental beyond the elementary level. Indeed (Section 7) the whole of I is provably not elementary, and G does not enter it.

5 New closed-form integrals from elementary χ_2

Conversely, (9) feeds the master formula to produce integrals that *do* close. The cleanest cross integral uses $u_2 = \phi^{-1}$, $u_{2/\sqrt{5}} = \phi^{-2}$, so $u_2 u_{2/\sqrt{5}} = \phi^{-3}$:

$$\boxed{\int_0^{\pi/2} \arctan(2 \sin x) \arctan\left(\frac{2}{\sqrt{5}} \sin x\right) dx = \frac{\pi^3}{24} - \frac{3\pi}{4} \ln^2 \phi} \quad (= 0.7463164407 \dots). \quad (10)$$

The diagonal $\int_0^{\pi/2} \arctan^2(\lambda \sin x) dx = \pi \chi_2(u_\lambda^2)$ closes at three clean λ , since $u_{\sqrt{\phi}}^2 = \phi^{-3}$, $u_{\sqrt{2+2\sqrt{2}}}^2 = \sqrt{2} - 1$, and $u_{2\phi^{3/2}}^2 = \phi^{-1}$:

$$\int_0^{\pi/2} \arctan^2(\sqrt{\phi} \sin x) dx = \frac{\pi^3}{24} - \frac{3\pi}{4} \ln^2 \phi, \quad \int_0^{\pi/2} \arctan^2(2\phi^{3/2} \sin x) dx = \frac{\pi^3}{12} - \frac{3\pi}{4} \ln^2 \phi. \quad (11)$$

The simplest non-elementary diagonal is the $(1, 1)$ case $\int_0^{\pi/2} \arctan^2(\sin x) dx = \pi \chi_2(3 - 2\sqrt{2})$.

6 A sibling trichotomy

The master formula has two siblings under the companion half-angle substitution $v_p = \frac{1-\sqrt{1-p^2}}{p} = \tanh(\frac{1}{2} \operatorname{arctanh} p)$ (for $|p| < 1$). Replacing one or both arctangents by $\operatorname{arctanh}$ produces:

$$\int_0^{\pi/2} \begin{cases} \arctan(p \sin x) \arctan(q \sin x) \\ \operatorname{arctanh}(p \sin x) \operatorname{arctanh}(q \sin x) \\ \arctan(p \sin x) \operatorname{arctanh}(q \sin x) \end{cases} dx = \begin{cases} \pi \chi_2(u_p u_q), \\ \pi \chi_2(v_p v_q), \\ \pi \operatorname{Ti}_2(u_p v_q), \end{cases} \quad (12)$$

where $\operatorname{Ti}_2(z) = \Im \operatorname{Li}_2(iz) = \int_0^z \frac{\operatorname{arctan} t}{t} dt$ is the inverse-tangent integral: like pairs give the Legendre chi, the mixed pair gives the inverse-tangent integral. The linear shadows are $\int_0^{\pi/2} \arctan(\lambda \sin x) dx = 2\chi_2(u_\lambda)$ (Reshetnikov) and, with $\theta = \arcsin \lambda$,

$$\int_0^{\pi/2} \operatorname{arctanh}(\lambda \sin x) dx = \theta \ln \tan \frac{\theta}{2} + 2 \operatorname{Cl}_2(\theta) - \frac{1}{2} \operatorname{Cl}_2(2\theta), \quad (13)$$

the latter showing that the $\operatorname{arctanh}$ branch is *not* a clean $2\chi_2(v_\lambda)$ mirror—that coincidence is special to $\arctan$. By $x \mapsto \frac{\pi}{2} - x$ the cos versions of all of these equal the sin versions. The weight-3 analogue $\int_0^{\pi/2} \arctan(p \sin x) \arctan(q \sin x) / \sin x dx$ has *no* single- χ_3 closed form: its coefficient $\frac{4^N}{(2N+1)\binom{2N}{N}}$ couples the two summation indices only through $N = m + n$, so it does not factor through $u_p u_q$ (numerically no clean χ_3 relation appears up to height 10^8).

7 An alternate form, and minimality

Landen-mirror form. Applying the χ_2 Landen law

$$\chi_2(x) + \chi_2\left(\frac{1-x}{1+x}\right) = \frac{\pi^2}{8} - \frac{1}{2} \ln x \ln \frac{1-x}{1+x} \quad (0 < x < 1) \quad (14)$$

to the three arguments of (4) sends $(\sqrt{2}-1)^2 \mapsto \frac{1}{\sqrt{2}}$, $(\sqrt{2}-1)(\sqrt{5}-2) \mapsto \sqrt{5}-\sqrt{2}$, $(\sqrt{5}-2)^2 \mapsto \frac{2}{\sqrt{5}}$, giving a second three-term representation

$$\frac{I}{\pi} = \mathcal{E} - \left[\chi_2\left(\frac{1}{\sqrt{2}}\right) + 2 \chi_2(\sqrt{5}-\sqrt{2}) + \chi_2\left(\frac{2}{\sqrt{5}}\right) \right], \quad (15)$$

with the elementary constant $\mathcal{E} = \frac{\pi^2}{2} - \frac{1}{2} \ln 2 \ln(1+\sqrt{2}) + 3 \ln 2 \ln \phi - \frac{3}{2} \ln 5 \ln \phi + (\ln(1+\sqrt{2}) + 3 \ln \phi) \ln(\sqrt{5}-\sqrt{2})$. The three mirror arguments are themselves non-elementary, so (15) is also irreducibly three-term; it trades the small original arguments for larger ones plus a richer logarithmic constant (the unit $\ln(\sqrt{5}-\sqrt{2}) = \ln 3 - \ln(\sqrt{5}+\sqrt{2})$ enters intrinsically).

Minimality. A height-bounded, canary-clean search ($\text{dps} \geq 120$, residual gate $\leq 10^{-105}$) finds no shorter form: I/π is *not* a rational multiple of any single $\chi_2(w)$ or $\text{Li}_2(w)$ over 209 algebraic arguments in $\mathbb{Q}(\sqrt{2}, \sqrt{5}, \sqrt{10})$; it is *not* elementary; it admits *no* two- χ_2 -plus-elementary representation; and Catalan’s constant G does not occur. The three-term χ_2 structure is minimal.

8 Verification and reproducibility

All displayed identities are verified in two independent engines—`mpmath` (`mp.quad`, `mp.pslq`) and Wolfram (`NIntegrate`, `FullSimplify`)—at the precisions noted: (4)/(2)/(3) exact to 160 digits; the master formula (5) and its differentiation proof symbolically (`FullSimplify`→0) and numerically at eight (p, q) ($\sim 10^{-48}$); the elementary values (9) symbolically and to $\sim 10^{-78}$; the new integrals (10)–(11) to $\sim 10^{-54}$; the sibling trichotomy (12) and the linear forms to $\sim 10^{-40}$; the Landen-mirror form (15) to $\sim 10^{-71}$; the classification and minimality searches to ≥ 130 digits with a Champernowne canary. Closed forms are in `src/PolyLog/identities/cleo-arctan-sin-chi2.wl`; scripts under `src/PolyLog/tools/scratch/cleo-chi2/`.

Attribution. The master formula (5) is Lee’s (a/562492, 2013), seeded by Reshetnikov’s single-arctangent ladder; the evaluation (2)/(3) is Cleo’s (a/570815) and Lee’s (a/571079); the elementary values (9) are classical (Lewin 1958; Campbell 2022). New here are the elementary- χ_2 classification (Proposition 2), the minimality of the three-term form, the Landen-mirror form (15), and the sibling trichotomy (12).